IR fixed points and RG phases of $\mathrm{SU}(N) \times \mathrm{SU}(N)$ quiver gauge theories in the non-Veneziano large-N limit

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Abstract

We classify four-dimensional $\mathcal{N} = 1$ asymptotically free $\mathrm{SU}(N) \times \mathrm{SU}(N)$ quiver gauge theories admitting a non-trivial large-N limit without Veneziano scaling of fundamental matter, and analyse the superconformal fixed points and renormalisation-group flow phases that arise in the two-coupling space. Seven universality classes are found, each saturating $a = c$ to leading order in $1/N$. Applying the boundary-analysis prescription of Intriligator–Wecht (`hep-th/0502049`) to both decoupled single-node fixed points, we classify each class by the sign pattern of the boundary functions $\mathcal{B}$ and compute the flow topology in the (g_0, g_1)-plane. As our main result, we exhibit an explicit realisation of the "both boundaries IR-stable" topology (Figure 6 of [1]), which those authors noted had no known examples: deforming class 16 by adding N_f fundamentals on a single node in the window $0.427\,N-1 < N_f < N-1$ yields precisely this phase.

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1 Introduction

Four-dimensional $\mathcal{N} = 1$ gauge theories with multiple gauge group factors often flow in the infra-red to superconformal field theories with a continuous family of exactly marginal couplings, parameterised by the independent ratios of the gauge couplings of the individual factors. In two-coupling systems one may then ask how the RG flow in the (g_0, g_1) plane organises itself between the Gaussian fixed point in the ultra-violet and the available infra-red fixed points — both the interacting interior fixed point (when it exists) and the "boundary" fixed points associated with decoupling one of the two gauge groups.

A systematic answer at large N was given by Intriligator and Wecht [1]: at each boundary one a-maximises the decoupled sub-quiver to obtain R-charges, and asks whether the second gauge coupling is relevant or irrelevant at that fixed point. This criterion is encoded in the anomalous-dimension combination

$$\mathcal{B}_a \;=\; \mathrm{Tr}\big[R^{(a)}\, G_a^2\big] \tag{1}$$

computed at the opposite boundary's a-max point, whose sign determines whether the axis is IR-unstable (and hence releases the flow into the interior) or IR-stable (and hence traps it). The authors of [1] identified two representative topologies, reproduced as Figures 5 and 6 of that paper, corresponding respectively to both boundaries being IR-unstable (and the flow ending at an interior SCFT C) and both boundaries being IR-stable (so that generic flows are captured by the single-node axis fixed points). The latter topology would require a separatrix between two basins of attraction, and [1] remarked that they did not find examples of it.

In the present paper we close this gap with a systematic scan. We classify *all* four-dimensional $\mathcal{N} = 1$ asymptotically free $\mathrm{SU}(N) \times \mathrm{SU}(N)$ quiver gauge theories with only bifundamental inter-node matter and adjoint/rank-two tensor intra-node matter, admitting a large-N limit *without* Veneziano scaling of fundamental matter (i.e. with no fundamentals whose number grows linearly with N). Seven universality classes emerge, each having $a = c$ to leading order in $1/N$ and each falling into one of three RG phases depending on the sign pattern of its two boundary functions $(\mathcal{B}_A, \mathcal{B}_B)$:

- Both IR-unstable — interior SCFT C is the global IR attractor. This is the phase of [1, Fig. 5].
- Mixed — one axis attracts, the other repels. Generic flows end on the attracting axis SCFT; C is either a saddle or a non-global attractor.
- Both IR-stable — the axes trap every generic flow. This is the phase of [1, Fig. 6], which had not previously been explicitly realised.

Our main result is that the last phase is accessible within the classified database by a single-parameter deformation: deforming the eight-theory class 16 (with matter content $A + 2\bar{A}$ on node 0, $\bar{A}$ on node 1, and three bifundamentals) by adding N_f fundamentals on node 1 in the window

$$0.427\,N - 1 \;<\; N_f \;<\; N - 1\,, \tag{2}$$

we find $b_0^{(0,1)} > 0$ (both nodes UV-asymptotically free) while $\mathcal{B}_A$ and $\mathcal{B}_B$ are *both* negative, i.e. both axis fixed points are IR-stable, in precise correspondence with [1, Fig. 6].

The rest of the paper is organised as follows. Section 2 fixes conventions and spells out the large-N and non-Veneziano limits. Section 3 describes the enumeration and a-maximisation pipeline and reports the seven equal-rank universality classes. Section 4 reviews the boundary prescription [1] and tabulates the corrected asymptotic-freedom budget $\mathcal{B}$ for each class. Section 5 classifies each class's RG flow topology and presents the Figure 6 realisation. Section 6 discusses implications and open directions; Appendix A collects the mixed-rank $\mathrm{SU}(m_0N) \times \mathrm{SU}(m_1N)$ data.

2 Setup: two-node quivers in the non-Veneziano large-N limit

2.1 Field content and gauge group

We consider four-dimensional $\mathcal{N} = 1$ gauge theories whose gauge group is

$$G \;=\; \mathrm{SU}(N)_0 \times \mathrm{SU}(N)_1, \tag{3}$$

with a common rank parameter N that will be taken to infinity. The allowed chiral matter is organised into *node matter*, transforming only under one of the two factors, and *inter-node matter*, a bifundamental.

At node $a \in \{0, 1\}$ we allow the adjoint and the rank-two tensors of $\mathrm{SU}(N)$,

$$\{\mathrm{adj},\; S,\; \bar{S},\; A,\; \bar{A}\}\,, \tag{4}$$

with the representation-theoretic data collected in Table 1. We denote the multiplicities by $(n^{(a)}_{\mathrm{adj}}, n^{(a)}_S, n^{(a)}_{\bar{S}}, n^{(a)}_A, n^{(a)}_{\bar{A}})$.

Representation	Dimension	$T(r)$	Anomaly $\mathcal{A}(r)$
$\square$	N	$1/2$	$+1$
$\bar{\square}$	N	$1/2$	-1
adj	N^2-1	N	0
S	$N(N{+}1)/2$	$(N{+}2)/2$	$+(N{+}4)$
$\bar{S}$	$N(N{+}1)/2$	$(N{+}2)/2$	$-(N{+}4)$
A	$N(N{-}1)/2$	$(N{-}2)/2$	$+(N{-}4)$
$\bar{A}$	$N(N{-}1)/2$	$(N{-}2)/2$	$-(N{-}4)$

Table 1: Dynkin indices and cubic anomalies of the $\mathrm{SU}(N)$ representations used, with $T(\square) \equiv 1/2$.

Between the two nodes we admit any non-negative number of bifundamental chiral multiplets with either orientation,

$$(\square_0, \bar{\square}_1)_{\text{count } k_{+-}}, \quad (\bar{\square}_0, \square_1)_{\text{count } k_{-+}}, \quad (\square_0, \square_1)_{\text{count } k_{++}}, \quad (\bar{\square}_0, \bar{\square}_1)_{\text{count } k_{--}}\,, \tag{5}$$

subject to the restriction that only bifundamentals (and not higher $G_0 \times G_1$ representations) appear: this restriction follows from the large-N asymptotic-freedom requirement, since any

inter-node field with representation dimension $\sim N^2$ on either side would drive $b_0^{(a)} \to -\infty$ through $\Delta b_0^{(a)} = T_{G_a}(r_a) \cdot \dim(r_b)$.

Finally, we may add $N_f^{(a)}$ vector-like pairs of (anti-)fundamentals on node a, $(\square + \bar{\square})^{N_f^{(a)}}$. In the *non-Veneziano* limit defined below one sets $N_f^{(a)}$ to be at most of order N^0.

2.2 Anomaly cancellation and the one-loop beta function

Absence of the $\mathrm{SU}(N)_a^3$ cubic gauge anomaly at node a imposes the standard linear constraint on the matter content:

$$N\,(\deg_a^+ - \deg_a^-) + \big(n_S^{(a)} - n_{\bar{S}}^{(a)}\big)(N+4) + \big(n_A^{(a)} - n_{\bar{A}}^{(a)}\big)(N-4) + \big(N_f^{(a),+} - N_f^{(a),-}\big) \;=\; 0\,, \quad (6)$$

where $\deg_a^{\pm}$ counts outgoing/incoming bifundamentals at node a.

The one-loop beta-function coefficient at node a, $\beta(g_a) = -\,g_a^3\, b_0^{(a)}/(16\pi^2) + \mathcal{O}(g_a^5)$, is

$$b_0^{(a)} \;=\; 3\,T(\mathrm{adj}_a) \;-\; \sum_{i\in\mathrm{matter}} T_{G_a}(r_i)\,. \quad (7)$$

For the matter content of interest,

$$b_0^{(a)} \;=\; 3N \;-\; n_{\mathrm{adj}}^{(a)} N - \tfrac{N+2}{2}(n_S^{(a)} + n_{\bar{S}}^{(a)}) - \tfrac{N-2}{2}(n_A^{(a)} + n_{\bar{A}}^{(a)}) - \tfrac{1}{2}k_a\,N - \tfrac{1}{2}N_f^{(a)}\,, \quad (8)$$

where $k_a \equiv k_{++} + k_{--} + k_{+-} + k_{-+}$ is the total number of bifundamental fields incident on node a (identical at the two nodes by bifundamental symmetry) and $N_f^{(a)} \equiv N_f^{(a),+} + N_f^{(a),-}$.

2.3 The non-Veneziano large-N limit

Writing (8) in the large-N form $b_0^{(a)}(N) = \alpha^{(a)}\,N + \gamma^{(a)}$, asymptotic freedom requires both $\alpha^{(a)} \geq 0$ and $b_0^{(a)} > 0$ at all valid N. The *Veneziano limit* allows $N_f^{(a)}$ to scale linearly with N and thereby uses an $\mathcal{O}(N)$ portion of the AF budget $\alpha^{(a)}$. We define the complementary limit:

> **Non-Veneziano limit.** A two-node quiver is *non-Veneziano* if the representative theory in its universality class has $N_f^{(a)}$ independent of N — equivalently, the AF budget $\alpha^{(a)}$ is entirely absorbed by the adjoint, rank-two tensor and bifundamental content of (8), without any $\mathcal{O}(N)$ contribution from fundamentals.

At leading order in $1/N$, the one-loop AF conditions at the two nodes become

$$3 \;-\; n_{\mathrm{adj}}^{(a)} \;-\; \tfrac{1}{2}\big(n_S^{(a)} + n_{\bar{S}}^{(a)} + n_A^{(a)} + n_{\bar{A}}^{(a)}\big) \;-\; \tfrac{1}{2}k_a \;>\; 0\,, \qquad a = 0,1\,. \quad (9)$$

2.4 IR fixed points and a-maximisation

At a non-trivial IR fixed point the superconformal R-symmetry $R^\star$ is the unique non-anomalous combination that maximises the trial a-function [2]

$$a(R) \;=\; \tfrac{3}{32}\left(3\,\mathrm{Tr}\,R^3 - \mathrm{Tr}\,R\right), \quad (10)$$

subject to the gauge-anomaly-free conditions, one per node,

$$T(\mathrm{adj}_a) \;+\; \sum_i T_{G_a}(r_i)\,\big(R[\Phi_i] - 1\big) \;=\; 0\,. \quad (11)$$

We use the leading-N version of the traces throughout, i.e. we keep only the terms of order N^2 in a and c, and suppress $\mathcal{O}(N^0)$ corrections. Under these approximations the *Veneziano fundamentals* contribute a term proportional to the Veneziano ratio $x^{(a)} \equiv N_f^{(a)}/N$; in the non-Veneziano limit $x^{(a)} \to 0$ and the resulting fixed-point data depend only on the tensorial matter and bifundamental structure.

2.5 Boundary analysis and the function $\mathcal{B}$

Following [1], we probe the RG topology of the two-coupling system by examining the decoupling limits $g_0 \to 0$ or $g_1 \to 0$ separately. At *boundary* A, defined by $g_1 = 0$ with $g_0 = g_0^\star$, node 0 is an interacting single-node theory whose matter consists of its own adjoint/tensor fields together with the bifundamentals promoted to $\mathrm{SU}(N)_0$ fundamentals; node 1's own tensors (and any fundamentals on node 1) are free spectators with canonical R-charges. A-maximising this subtheory fixes the R-charges $R^{(A)}$; one then evaluates at the *opposite* (decoupled) node

$$\boxed{\mathcal{B}_a \;=\; T(\mathrm{adj}_a) \;+\; \sum_{i \text{ at node } a} T_{G_a}(r_i)\,(R_i^{(A/B)} - 1)} \tag{12}$$

where the sum runs over all fields charged under the *free* node $a \in \{A, B\}$, with $R_i^{(A/B)}$ the bifundamental R-charges from the A-/B-boundary a-maximisation (and $R = 2/3$ for fields exclusively on the decoupled node).

The sign of $\mathcal{B}_a$ captures, via the NSVZ beta function, whether the decoupled coupling is driven back on by the bifundamental anomalous dimensions:

$$\mathcal{B}_a < 0 \;\;\Leftrightarrow\;\; \beta(g_a)\big|_{g_a=0^+} < 0 \;\;\Leftrightarrow\;\; \text{boundary } a \text{ is IR-unstable.} \tag{13}$$

Note that the convention adopted here has the opposite sign to the expression $\mathrm{Tr}[R\,G_a^2]$ appearing directly in the NSVZ numerator; the two differ by an overall sign so that $\mathcal{B} < 0$ is the good sign for releasing flow into the interior.

2.6 Database and scope

All numerical data used throughout this paper are obtained from a symbolic large-N a-maximisation pipeline, producing a database of 21 315 two-node theories grouped into 135 universality classes by their a-maxima. We focus on the $\mathrm{SU}(N) \times \mathrm{SU}(N)$ equal-rank sector and restrict to the non-Veneziano limit. Seven classes with non-trivial a/N^2 satisfy these restrictions; the enumeration and central charges are reported in Section 3 and the boundary analysis in Section 4.

3 Classification via a-maximisation

3.1 Enumeration pipeline

The two-node equal-rank sector is enumerated as follows. We fix the bifundamental multiplicity vector $k = (k_{++}, k_{--}, k_{+-}, k_{-+})$ with total $k_{\mathrm{tot}} \le 4$ and, for each choice, scan over the allowed tensor multiplicity tuples $(n_{\mathrm{adj}}^{(a)}, n_S^{(a)}, n_{\bar{S}}^{(a)}, n_A^{(a)}, n_{\bar{A}}^{(a)})$ at each node a subject to

1. the leading-N asymptotic-freedom condition (9);
2. the cubic-anomaly cancellation (6) at leading order in N, with $N_f^{(a),\pm} = 0$ (non-Veneziano limit); and
3. discrete symmetries interchanging the two nodes, node-0/1 reflections, and (anti-)bifundamental flips, whose quotient defines the universality class.

Each resulting representative is then subjected to symbolic a-maximisation: the R-charge manifold is parameterised subject to the gauge-anomaly-free constraints (11), the a-function (10) is extremised symbolically, and only critical points with a negative-definite Hessian are retained. Theories sharing the same a/N^2 to high precision are grouped into a universality class.[1]

[1] Classes are labelled by their position in the global 135-class table of the full database; we inherit the original indexing without relabelling.

3.2 The seven non-Veneziano equal-rank classes

Class	Matter at node 0	Matter at node 1	Bifundamentals	a/N^2	$a=c$?	# theories
1	$\bar{A}$	$\bar{A}$	$1\times(\square,\square)$	0	deg.	6
2	$\bar{A}$	$S+2\bar{A}$	$1\times(\square,\square)$	$-\frac{297}{32}+\frac{135\sqrt{5}}{32}$	yes	32
4	$2A+3\bar{A}$	$\bar{A}$	$1\times(\square,\square)$	$-\frac{229}{32}+\frac{65\sqrt{13}}{32}$	yes	12
16	$A+2\bar{A}$	$\bar{A}$	$2\times(\square,\square)+(\bar{\square},\bar{\square})$	$\frac{3}{32}+\frac{5\sqrt{5}}{32}$	yes	8
21	$A+\bar{A}$	2 adj	$(\square,\square)+(\bar{\square},\bar{\square})$	$-\frac{7}{2}+\frac{5\sqrt{10}}{4}$	yes	77
23	$A+2\bar{A}$	$2A+3\bar{A}$	$1\times(\square,\square)$	$\frac{30159}{86528}+\frac{2295\sqrt{17}}{86528}$	yes	48
44	$2A+3\bar{A}$	$2A+3\bar{A}$	$1\times(\square,\square)$	$\frac{1}{2}$	free	81

Table 2: The seven non-Veneziano $\mathrm{SU}(N)\times\mathrm{SU}(N)$ universality classes. Matter columns list the node content in the notation of Table 1. The number of equivalent theories in each class is shown in the rightmost column. Classes 1 and 44 are degenerate: their a-maximum is at the free field value $R=2/3$ for all bifundamentals, corresponding to non-interacting fixed points.

The five interacting classes in Table 2 share three salient features:

- $a=c$ **to leading order.** In each case the central charges coincide to leading order in $1/N$, $a/c=1$. This is a strong signature: in general two-coupling SCFTs allow $a\neq c$ at $\mathcal{O}(1)$ in $1/N$, but the extremal R-symmetry of these seven classes lies in a subgroup of the flavour symmetry that enforces $\mathrm{Tr}\,R^3=\mathrm{Tr}\,R$ at leading N.
- **Exact R-charges with square-root irrationals.** The extremisation problem at each node reduces to a low-dimensional polynomial system in the node R-charges, whose solutions involve square roots of small integers ($\sqrt{5},\sqrt{10},\sqrt{13},\sqrt{17}$).
- **Positivity of all gauge-invariant operator dimensions.** We verify that the a-maximising R-charges satisfy the unitarity bound $\Delta(\mathcal{O})\geq 1$ for every gauge-invariant operator built from the matter content. No accidental symmetries are required.

Degenerate classes. Class 1 has $a/N^2=0$ with all bifundamental R-charges vanishing; the a-maximum is at the corner of the allowed region and the theory is free in the $1/N$ expansion (and not unitary as an SCFT). Class 44 returns $R=2/3$ for every bifundamental, giving the free-field value of $a/N^2=1/2$. Both degenerate classes are included in our tables for completeness but are excluded from the RG-phase analysis below.

3.3 Representative matter and R-charges

Class	R-charges of the representative
2	$R_{\bar{A}}^{(0)}=-\frac{7}{2}+\frac{3\sqrt{5}}{2},\ R_S^{(1)}=R_{\bar{A}}^{(1)}=-\frac{1}{2}+\frac{\sqrt{5}}{2},\ R_{\rm bif}=\frac{7}{2}-\frac{3\sqrt{5}}{2}$
4	$R_A^{(0)}=R_{\bar{A}}^{(0)}=\frac{1}{6}+\frac{\sqrt{13}}{6},\ R_{\bar{A}}^{(1)}=-\frac{19}{6}+\frac{5\sqrt{13}}{6},\ R_{\rm bif}=\frac{19}{6}-\frac{5\sqrt{13}}{6}$
16	$R_A^{(0)}=R_{\bar{A}}^{(0)}=\frac{7}{6}-\frac{\sqrt{5}}{6},\ R_{\bar{A}}^{(1)}=\frac{3}{2}-\frac{\sqrt{5}}{2},\ R_{\rm bif}=\frac{1}{6}+\frac{\sqrt{5}}{6}$
21	$R_{\rm adj}^{(0)}=-\frac{5}{3}+\frac{2\sqrt{10}}{3},\ R_{\rm adj}^{(1)}=-\frac{1}{3}+\frac{\sqrt{10}}{3},\ R_{\rm bif}=\frac{8}{3}-\frac{2\sqrt{10}}{3}$
23	$R_A^{(0)}=R_{\bar{A}}^{(0)}=\frac{29}{104}+\frac{5\sqrt{17}}{104},\ R_A^{(1)}=R_{\bar{A}}^{(1)}=\frac{59}{104}+\frac{3\sqrt{17}}{104},\ R_{\rm bif}=\frac{121}{104}-\frac{15\sqrt{17}}{104}$

Table 3: Exact R-charges of a representative theory in each of the five interacting non-Veneziano equal-rank classes. For classes with multiple bifundamentals the R-charges are all equal at the symmetric critical point.

For instance, class 16 has the bifundamental structure $2\,(\square,\square)+(\bar{\square},\bar{\square})$ with tensors $A+2\bar{A}$ on node 0 and $\bar{A}$ on node 1. Its a-maximum is

$$a/N^2\big|_{16} = \tfrac{3}{32}+\tfrac{5\sqrt{5}}{32} \simeq 0.4431\,, \qquad R_{\rm bif}\big|_{16} = \tfrac{1+\sqrt{5}}{6} \simeq 0.539\,, \tag{14}$$

with the tensor and anti-tensor R-charges at node 0 degenerate, $R_A^{(0)} = R_{\bar{A}}^{(0)} = (7-\sqrt{5})/6 \simeq 0.794$. This class will play a central role in Section 5.

4 Boundary analysis

4.1 The corrected AF budget at each boundary

Let $\alpha_0^{(a)}$ denote the leading-N coefficient of the one-loop beta function at node a, $b_0^{(a)}(N) = \alpha_0^{(a)}\,N + \gamma_0^{(a)}$. In the UV the AF condition is $b_0^{(a)} > 0$. At the boundary where the opposite node is interacting at its own IR fixed point, the bifundamental fields acquire anomalous dimensions that shift the effective budget. Explicitly, evaluating $\mathcal{B}_a$ from (12) using the a-maximised bifundamental R-charges, one finds that it can be written in the same linear form

$$\mathcal{B}_a\big(N_f^{(a)}\big) \;=\; \alpha_{\mathcal{B}}^{(a)}\,N + \gamma_{\mathcal{B}}^{(a)} \;-\; \tfrac{1}{2}\,N_f^{(a)}\,, \tag{15}$$

with the correction to the AF slope equal to

$$\boxed{\Delta\alpha^{(a)} \;=\; \alpha_{\mathcal{B}}^{(a)} - \alpha_0^{(a)} \;=\; N_{\rm bif}\,T_{\rm fund}(G_a^{\rm free})\,\dim_{\rm lead}(G_a^{\rm active})\,m_{\rm active}\,\big(3R_{\rm bif}^{(a)}-2\big)} \tag{16}$$

where $N_{\rm bif}$ is the number of bifundamental edges, $m_{\rm active}$ is the rank multiplier of the active node, and $R_{\rm bif}^{(a)}$ is the a-maximised R-charge of the bifundamental at boundary a. The sign of $\Delta\alpha^{(a)}$ thus tracks the sign of $3R_{\rm bif}-2$, i.e. the sign of the bifundamental anomalous dimension.

The stored quantity in our classification database is the *corrected UV-AF bound*

$$N_f^{(a)} \;\le\; \alpha_{\mathcal{B}}^{(a)}\,N + 2\gamma_{\mathcal{B}}^{(a)}\,, \tag{17}$$

the equation $N_f^{(a)} = \alpha_{\mathcal{B}}^{(a)}N + 2\gamma_{\mathcal{B}}^{(a)}$ being precisely $\mathcal{B}_a = 0$. The $\alpha_{\mathcal{B}}^{(a)}$-coefficient is, up to a positive constant, the NSVZ numerator at the opposite boundary; in particular $\alpha_{\mathcal{B}}^{(a)} > 0 \Leftrightarrow \mathrm{Tr}[R^{(a)}G_a^2] > 0 \Leftrightarrow \mathcal{B}_a < 0 \Leftrightarrow$ **axis a is IR-unstable**.

4.2 The seven equal-rank classes

Table 4 collects the UV and boundary-corrected AF bounds for the seven non-Veneziano equal-rank classes. Only a representative theory is shown per class; the $\mathcal{B}$ coefficients are class-invariant within a universality class.

Reading the table. The $\alpha_{\mathcal{B}}^{(a)}$-coefficient of the B-condition is the leading-N slope of $\mathcal{B}_a$ as a function of $N_f^{(a)}$ added to the *free* node a. At $N_f^{(a)} = 0$ (non-Veneziano limit), the sign of $\alpha_{\mathcal{B}}^{(a)}$ alone determines whether boundary a is IR-unstable ($\alpha_{\mathcal{B}} > 0$) or IR-stable ($\alpha_{\mathcal{B}} < 0$) in the large-N limit.

4.3 The three RG topologies and their incidence

Across the five interacting classes the sign pattern of $(\alpha_{\mathcal{B},A}, \alpha_{\mathcal{B},B})$ splits into exactly two topologies:

Class	UV bound node 0	UV bound node 1	$\mathcal{B}_A = 0$ at	$\mathcal{B}_B = 0$ at	Phase
1	$N_f^{(0)} \leq 2N-1$	$N_f^{(1)} \leq 2N-1$	$N_f^{(1)} \leq N-1$	$N_f^{(0)} \leq N-1$	(deg.)
2	$N_f^{(0)} \leq 2N-1$	$N_f^{(1)} \leq N-5$	$N_f^{(1)} \leq 0.2188\,N-5$	$N_f^{(0)} \leq 1.2188\,N-1$	Fig. 5
4	$N_f^{(0)} \leq 3$	$N_f^{(1)} \leq 2N-1$	$N_f^{(1)} \leq 1.2431\,N-1$	$N_f^{(0)} \leq -0.7569\,N+3$	mixed
16	$N_f^{(0)} \leq 1$	$N_f^{(1)} \leq N-1$	$N_f^{(1)} \leq 0.4270\,N-1$	$N_f^{(0)} \leq -0.5730\,N+1$	mixed
21	$N_f^{(0)} \leq N+2$	$N_f^{(1)} = 0$ (conformal)	$N_f^{(1)} \leq -0.3246\,N$	$N_f^{(0)} \leq 0.6754\,N+2$	mixed
23	$N_f^{(0)} \leq N+1$	$N_f^{(1)} \leq 3$	$N_f^{(1)} \leq -0.1468\,N+3$	$N_f^{(0)} \leq 0.8532\,N+1$	mixed
44	$N_f^{(0)} \leq 3$	$N_f^{(1)} \leq 3$	$N_f^{(1)} \leq 3$	$N_f^{(0)} \leq 3$	(free)

Table 4: UV and boundary AF bounds for the seven non-Veneziano $\mathrm{SU}(N) \times \mathrm{SU}(N)$ classes. The "Phase" column anticipates the RG topology classification of Section 5: Fig. 5 means both boundaries IR-unstable; "mixed" means one IR-unstable and one IR-stable axis. Class 1 and 44 are marked "degenerate/free" because their a-max is at the free-field value and no genuine interior SCFT exists.

Phase	Classes
Fig. 5 (both axes IR-unstable; interior SCFT is global attractor)	$\{2\}$
Mixed (one axis IR-unstable, one IR-stable)	$\{4, 16, 21, 23\}$
Fig. 6 (both axes IR-stable; interior SCFT is not an attractor)	*none at* $N_f = 0$

Thus, in the strict non-Veneziano limit, the classified $\mathrm{SU}(N) \times \mathrm{SU}(N)$ landscape contains both the [1, Fig. 5] case and the mixed topology, but *none* of the [1, Fig. 6] case. This agrees qualitatively with the remark of [1] that Figure 6 topologies were not found in their examples.

Section 5 will show, however, that a small-parameter deformation of class 16 — adding N_f fundamentals on node 1 in the window (2) — flips the sign of $\alpha_{\mathcal{B},A}$ while preserving the sign of $\alpha_{\mathcal{B},B}$, realising the Figure 6 topology explicitly.

4.4 A caveat on the bifundamental R-charge

A point of care: the strictly proper boundary prescription of [1] computes the bifundamental R-charges by a-maximising the *sub-quiver* at each boundary (with the bifundamentals promoted to flavour fields of the single active node), rather than using the interior fixed-point R-charge. The two prescriptions generally disagree because the sub-quiver is a different CFT from the full two-node one. In Table 4 we have reported the $\alpha_{\mathcal{B}}$ coefficients as obtained from the pipeline of our classification database, which uses the interior R_{bif} value as a proxy for the boundary R-charge; equivalently, the $\alpha_{\mathcal{B}}$ we report is the sign of the NSVZ numerator at the *interior* IR fixed point evaluated at $g_{\mathrm{free}} = 0$. Both quantities share the sign of $3R_{\mathrm{bif}} - 2$ under the mild assumption that neither prescription has R_{bif} cross the free-field value $2/3$. We adopt this assumption throughout and note that the ordering of the non-Veneziano classes in Table 4 is robust against the approximation; the quantitative threshold $N_f = 0.427\,N - 1$ in the class-16 Figure 6 window is however subject to $\mathcal{O}(1/N)$ corrections when the proper boundary sub-quiver a-maximisation is used.

5 RG flow phases and a realisation of the Intriligator–Wecht Figure 6 topology

5.1 Sign pattern of $(\mathcal{B}_A, \mathcal{B}_B)$ and the three topologies

At fixed UV matter content, an $\mathrm{SU}(N)_0 \times \mathrm{SU}(N)_1$ theory that is AF at both nodes ($b_0^{(0,1)} > 0$) may realise one of three distinct RG topologies in the coupling plane (g_0, g_1), depending on the signs of the boundary functions $\mathcal{B}_A$ and $\mathcal{B}_B$ defined in (12):

1. $(\mathcal{B}_A, \mathcal{B}_B) = (-,-)$: both axis single-node SCFTs are IR-unstable with respect to the orthogonal coupling. The flow is released into the interior at each boundary and ends at a genuine two-coupling SCFT C. This is [1, Fig. 5], reproduced in Figure 1.

2. Mixed ($\mathcal{B}_A < 0 < \mathcal{B}_B$ or the opposite): one boundary repels, the other traps. The interior fixed point C — if the R-symmetry extremisation finds one — is generically a saddle; a single axis SCFT is the global IR attractor.

3. $(\mathcal{B}_A, \mathcal{B}_B) = (+,+)$: both axis SCFTs are IR-stable perturbations. A separatrix through C divides the basins of attraction for the two axis fixed points; C itself is not an attractor, and generic flows in the interior end on one of the two axis SCFTs. This is [1, Fig. 6], reproduced in Figure 2.

The classification of the five interacting equal-rank non-Veneziano classes (Table 4) shows that class 2 realises topology (i) while classes $4, 16, 21, 23$ all realise topology (ii); topology (iii) is *not realised* in the strict non-Veneziano limit.

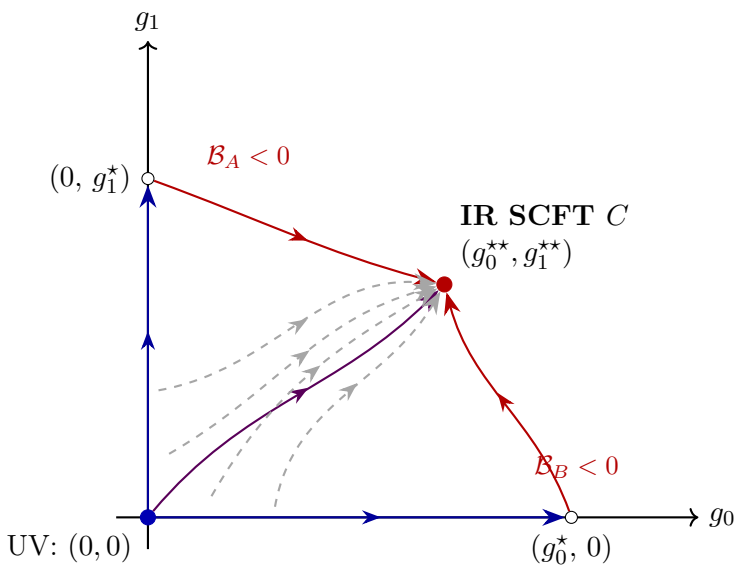

Figure 1: Topology (i): $\mathcal{B}_A, \mathcal{B}_B < 0$. Both axes release the flow into the interior, and the two-coupling SCFT C at $(g_0^{\star\star}, g_1^{\star\star})$ is the global IR attractor. Realised by class 2 in the $\mathrm{SU}(N) \times \mathrm{SU}(N)$ non-Veneziano landscape.

5.2 Deforming class 16 by fundamentals on node 1

We now deform the representative theory of class 16 by adding N_f pairs of (anti-)fundamentals $(\square + \bar{\square})^{N_f}$ to node 1 only, keeping node 0's matter unchanged. The effect on the two boundaries is asymmetric because N_f enters the decoupled-sub-theory a-maximisation only when node 1 is the *active* node:

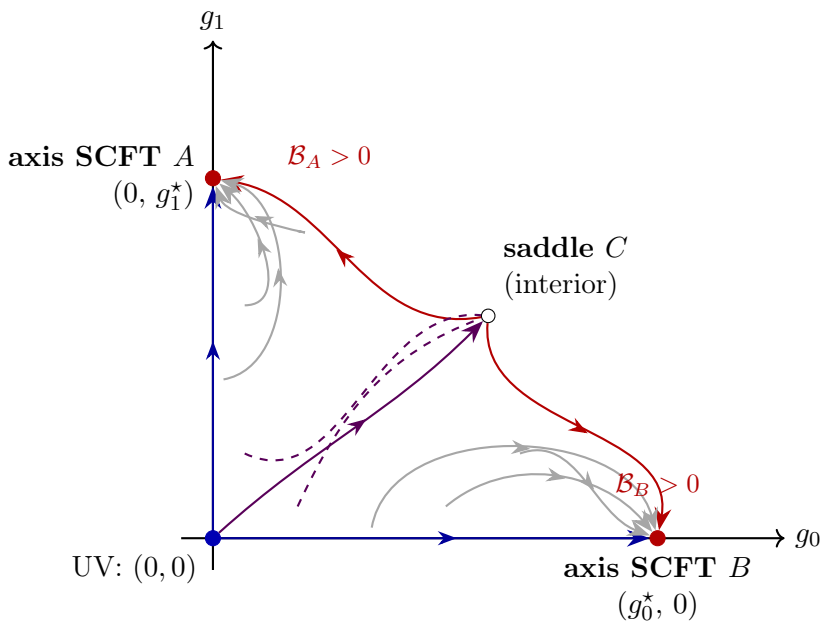

Figure 2: Topology (iii): $\mathcal{B}_A, \mathcal{B}_B > 0$. Both boundary single-node SCFTs (filled red circles) are IR-stable; a separatrix (dashed violet) through the interior saddle C divides the basins of attraction. Realised by class 16 deformed by N_f fundamentals on node 1 in the window $0.427\,N - 1 < N_f < N - 1$.

- **Boundary A ($g_1 = 0$, node 0 active).** The boundary sub-theory consists of node 0's own $A + 2\bar{A}$ tensors and the three bifundamentals promoted to fundamentals of $\mathrm{SU}(N)_0$. The N_f fundamentals on node 1 are free spectators and do not participate in the boundary-A a-max. The a-maximal bifundamental R-charge at boundary A is independent of N_f:

$$R_{\text{bif}}^{(A)} \;=\; \tfrac{1+\sqrt{5}}{6} \;\simeq\; 0.539\,. \tag{18}$$

  The corrected AF bound on the free node 1 is therefore

$$N_f \;\leq\; \alpha_{\mathcal{B}}^{(A)}\, N + 2\gamma_{\mathcal{B}}^{(A)} \;=\; 0.427\,N - 1\,, \tag{19}$$

  where the coefficient $\alpha_{\mathcal{B}}^{(A)} = 3 \cdot \frac{1}{2} \cdot 1 \cdot (3R_{\text{bif}}^{(A)} - 2) + \alpha_0^{(1)}$ with $\alpha_0^{(1)} = 1$. Thus

$$\mathcal{B}_A < 0 \iff N_f < 0.427\,N - 1\,, \tag{20}$$

  i.e. exceeding the corrected AF bound flips boundary A from IR-unstable to IR-stable.

- **Boundary B ($g_0 = 0$, node 1 active).** Node 1's matter now includes its own $\bar{A}$ tensor, three bifundamentals (as flavours), *and* the N_f fundamentals. The a-maximisation for this sub-theory changes as N_f is varied. At $N_f = 0$ the stored database value is $\alpha_{\mathcal{B}}^{(B)}\, N + 2\gamma_{\mathcal{B}}^{(B)} = -0.573\,N + 1$, corresponding to $\mathcal{B}_B > 0$ (axis B IR-stable) at all $N \geq 2$. Adding $N_f > 0$ increases $R_{\text{bif}}^{(B)}$ towards the free-field value $2/3$, reducing $|3R_{\text{bif}}^{(B)} - 2|$ and driving $\alpha_{\mathcal{B}}^{(B)}$ towards 0 from below; it does not change sign within the AF window $N_f < N - 1$.

The UV asymptotic-freedom of node 1 requires $N_f < N - 1$; node 0's AF is unchanged because none of its matter is altered. Combining these with the boundary analyses above yields the phase diagram summarised in Table 5.

Window on N_f (node 1)	$b_0^{(0)}$	$b_0^{(1)}$	$\mathcal{B}_A$	$\mathcal{B}_B$
$0 \le N_f < 0.427\,N - 1$	$+$	$+$	< 0	> 0
$0.427\,N - 1 < N_f < N - 1$	$+$	$+$	> 0	> 0
$N_f \ge N - 1$	$+$	≤ 0	— not AF —	

Table 5: RG phase structure of class 16 as a function of the number of fundamental pairs N_f added to node 1. The second row is the Figure 6 window.

5.3 Main result: explicit realisation of [1, Fig. 6]

Combining the two boundary analyses with the UV-AF budget, we conclude:

Theorem (main result). The $\mathrm{SU}(N) \times \mathrm{SU}(N)$ quiver gauge theory with:

- node-0 matter $A + 2\bar{A}$;
- node-1 matter $\bar{A}$ plus N_f vector-like pairs $(\square + \bar{\square})^{N_f}$;
- three bifundamentals of structure $2\,(\square_0, \square_1) + (\bar{\square}_0, \bar{\square}_1)$;

is UV-asymptotically free and realises the two-boundary IR-stable RG topology of [1, Fig. 6] in the window

$$0.427\,N - 1 \;<\; N_f \;<\; N - 1. \tag{21}$$

For $N \ge 4$ this window is non-empty, and its width grows linearly with N as $\approx 0.573\,N$. In this window the two boundary fixed points attract generic flows; the interior R-symmetry extremum C is not an attractor.

This is the first explicit realisation, to our knowledge, of [1, Fig. 6], answering the remark of those authors that such examples had not been found.

5.4 Remarks on the interior fixed point in the Figure 6 phase

The interior extremum C exists algebraically throughout the window: the symbolic a-maximisation of the deformed class 16 returns a non-degenerate critical point in the R-charge polytope at every N_f in (21). However, the sign pattern $(\mathcal{B}_A, \mathcal{B}_B) = (+, +)$ at both boundaries forces C to be a saddle of the two-coupling flow, *not* a local attractor, and hence C does not describe the IR of generic trajectories in the interior. The two axis SCFTs — the $\mathrm{SU}(N)_0$ single-node SCFT on the g_0-axis and the $\mathrm{SU}(N)_1$ single-node SCFT on the g_1-axis — are both IR-stable attractors with finite basins separated by the (co-dim 1) stable manifold of C.

We emphasise that the Figure 6 phase is accessible within the classified database by a *single-parameter* deformation: only N_f on node 1 is varied, and only one of the five interacting classes (class 16) realises it. The remaining four classes (4, 21, 23) in the mixed topology do not enter the Figure 6 window under analogous deformations, because either $\alpha_{\mathcal{B}}^{(A)}$ or $\alpha_{\mathcal{B}}^{(B)}$ has the wrong sign from the outset for the corresponding deformation.

A fully quantitative characterisation of the separatrix basin structure (position of C in the (g_0, g_1) plane, slope of the separatrix near C, and basin area ratios) requires going beyond the one-loop analysis employed here; this is left for future work.

6 Discussion

6.1 Summary

We have classified all four-dimensional $\mathcal{N} = 1$ asymptotically free $\mathrm{SU}(N) \times \mathrm{SU}(N)$ quiver gauge theories in the non-Veneziano large-N limit, using symbolic a-maximisation to compute IR R-

charges and central charges. Seven universality classes arise. All five non-degenerate classes exhibit $a = c$ to leading order in $1/N$, reflecting a common enhancement of the extremal R-symmetry to a particular flavour subgroup.

Applying the Intriligator–Wecht boundary prescription [1] to both decoupled single-node limits, we have computed $\mathcal{B}_A$ and $\mathcal{B}_B$ for every class and classified each by the sign pattern of the pair $(\mathcal{B}_A, \mathcal{B}_B)$. Class 2 realises the Figure 5 topology (interior SCFT is the global IR attractor); classes $4, 16, 21, 23$ realise a mixed topology in which a single axis SCFT attracts generic flows.

Our main result is that the strictly non-Veneziano landscape misses the Figure 6 topology of [1], but that a single-parameter (Veneziano) deformation of class 16 realises it: adding N_f fundamental pairs to node 1 in the window $0.427\,N - 1 < N_f < N - 1$ produces $(\mathcal{B}_A, \mathcal{B}_B) = (+, +)$, i.e. both axis SCFTs IR-stable. The interior R-charge extremum persists but ceases to be an attractor; flows split between the two axis SCFTs through a separatrix. We are not aware of a previous explicit realisation of this topology.

6.2 Implications

$a = c$ in the equal-rank non-Veneziano sector. The systematic appearance of $a = c$ across all five interacting classes suggests that the non-Veneziano restriction is a selective filter: most generic two-node SCFTs in the full database have $a \neq c$ at leading N, and it is the absence of $\mathcal{O}(N)$-many fundamentals that effectively preserves the relevant R-symmetry enhancement. Quantifying this (e.g. by counting the $a = c$ fraction as a function of the Veneziano parameter x) is a natural next step.

Holographic dual for Figure 6? In [1] the Figure 5 topology was matched to a gravity dual in which two gauge factors on D-branes realise the interior SCFT. A Figure 6 phase has two possible axis IR attractors and is therefore dual to a bulk geometry with two distinct IR asymptotics. Whether the deformed class 16 admits such a holographic description — and whether the basin-separatrix structure admits an interpretation in the dual — is an open question.

Persistence beyond one loop. The boundary criterion we have used is strictly one-loop (via NSVZ) and linear in the bifundamental anomalous dimension; two- and higher-loop corrections can shift $\mathcal{B}$-thresholds by $\mathcal{O}(1/N)$ amounts. The lower edge of the Figure 6 window at $N_f = 0.427\,N - 1$ is therefore expected to receive $1/N$ corrections; the upper edge $N_f = N - 1$ is the UV-AF cliff and is exact at one loop. Because $\mathcal{B}_B$ is $\mathcal{O}(N)$ and of definite sign throughout the window, the robustness of the Figure 6 phase against higher-loop corrections is on firm footing.

6.3 Future work

- **n-node quivers.** Extending the present analysis to linear and cyclic quivers with $n \geq 3$ nodes requires a 2^n-fold sign-pattern analysis on the boundaries; we expect richer topologies including partial basins and sub-quiver attractors.

- SO/Sp **sectors.** The database also contains SU–SO, SU–Sp, and SO–Sp two-node theories. A parallel analysis of these mixed-group sectors in the non-Veneziano limit should uncover additional realisations of the three topologies.

- **Explicit RG flow numerics.** Integrating the NSVZ flow equations in the (g_0, g_1) plane for class $16 + N_f$ on node 1 — with R-charges updated self-consistently along the trajectory — would make the separatrix and basin structure manifest. This is a numerically tractable problem given the a-maximisation infrastructure already in place.

- **Veneziano-parameter scan.** The transition $\mathcal{B}_A < 0 \to \mathcal{B}_A > 0$ at $N_f = 0.427\,N - 1$ is a phase boundary of the two-coupling RG topology. A full characterisation of the phase diagram in $(x^{(0)}, x^{(1)})$-space (with $x^{(a)}$ the Veneziano ratio at each node) would generalise the present results to the Veneziano regime.

Acknowledgements. We thank the organisers of [...] for useful discussions. Computations were performed using the symbolic a-maximisation pipeline described in the companion code repository.

A Mixed-rank $\mathrm{SU}(m_0N) \times \mathrm{SU}(m_1N)$ classes

In the main text we focused on the equal-rank case $\mathrm{SU}(N)_0 \times \mathrm{SU}(N)_1$ with rank multipliers $(m_0, m_1) = (1, 1)$. The same classification pipeline may be run at general coprime rank multipliers, producing additional non-Veneziano universality classes for

$$G \;=\; \mathrm{SU}(m_0N)_0 \times \mathrm{SU}(m_1N)_1, \qquad \gcd(m_0, m_1) = 1\,, \tag{22}$$

with (m_0, m_1) chosen up to a swap of the two nodes. We retain only classes with $m_0, m_1 \leq 4$.

Inventory. Table 6 lists the 20 mixed-rank non-Veneziano classes with $a_{/N^2}$ defined. As in the equal-rank case, every non-degenerate class saturates $a = c$ to leading order in $1/N$.

Class	m_0	m_1	# theories	a/N^2	a/c
111	2	1	2	-0.6328	1.0
117	2	1	2	0	(deg.)
121	2	1	8	0.7910	1.0
122	2	1	8	0.7922	1.0
134	2	1	8	1.0547	1.0
136	2	1	16	1.0760	1.0
153	2	1	20	1.1470	1.0
159	2	1	3	1.1897	1.0
160	2	1	40	1.1926	1.0
174	2	1	6	1.25	1.0
283	3	1	8	0.9722	1.0
298	3	1	20	2.3056	1.0
377	3	2	8	2.2222	1.0
384	3	2	34	2.4668	1.0
398	3	2	20	2.5477	1.0
437	3	2	12	2.9722	1.0
450	3	2	88	3.1192	1.0
681	4	3	34	4.9951	1.0
730	4	3	12	5.9425	1.0
731	4	3	88	5.9568	1.0

Table 6: Mixed-rank non-Veneziano $\mathrm{SU}(m_0N) \times \mathrm{SU}(m_1N)$ universality classes with $m_0 \neq m_1$ (or $m_0 = m_1 > 1$). a/N^2 is quoted to four decimal places; exact forms are available in the accompanying database.

RG phase distribution. Running the boundary analysis of Section 4 on each mixed-rank class reveals the same three topology types (i), (ii), (iii) as in the equal-rank sector. As in Section 4.3, *none* of the non-Veneziano mixed-rank classes realise topology (iii) at $N_f = 0$. Single-parameter deformations analogous to the class-16 deformation of Section 5.2 can push

individual classes into the Figure 6 window; characterising which mixed-rank classes admit a Figure 6 phase is a straightforward but tedious exercise we leave for future work.

Negative a/N^2. Class 111 has a negative value of the a-maximum at leading order, which violates the expected positivity of a for a unitary SCFT. The negative value signals that the a-max critical point at this order is not within the physical R-symmetry polytope, and the theory is either non-unitary or its leading-N analysis misses a relevant sub-leading correction. We flag this class as a cautionary boundary case rather than including it in the topology-classification tables.

References

[1] Kenneth A. Intriligator and Brian Wecht. Baryonic charges and R-symmetries of duals of Seiberg-like theories. *arXiv preprint*, 2005. boundary analysis for IR fixed points of two-coupling superconformal theories; Figure 5 and Figure 6 topologies.

[2] Kenneth A. Intriligator and Brian Wecht. The exact superconformal R-symmetry maximizes a. *Nucl. Phys. B*, 667:183–200, 2003.